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School Deen Dayal Upadhyaya University

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007

A-4 Spiral Note Book

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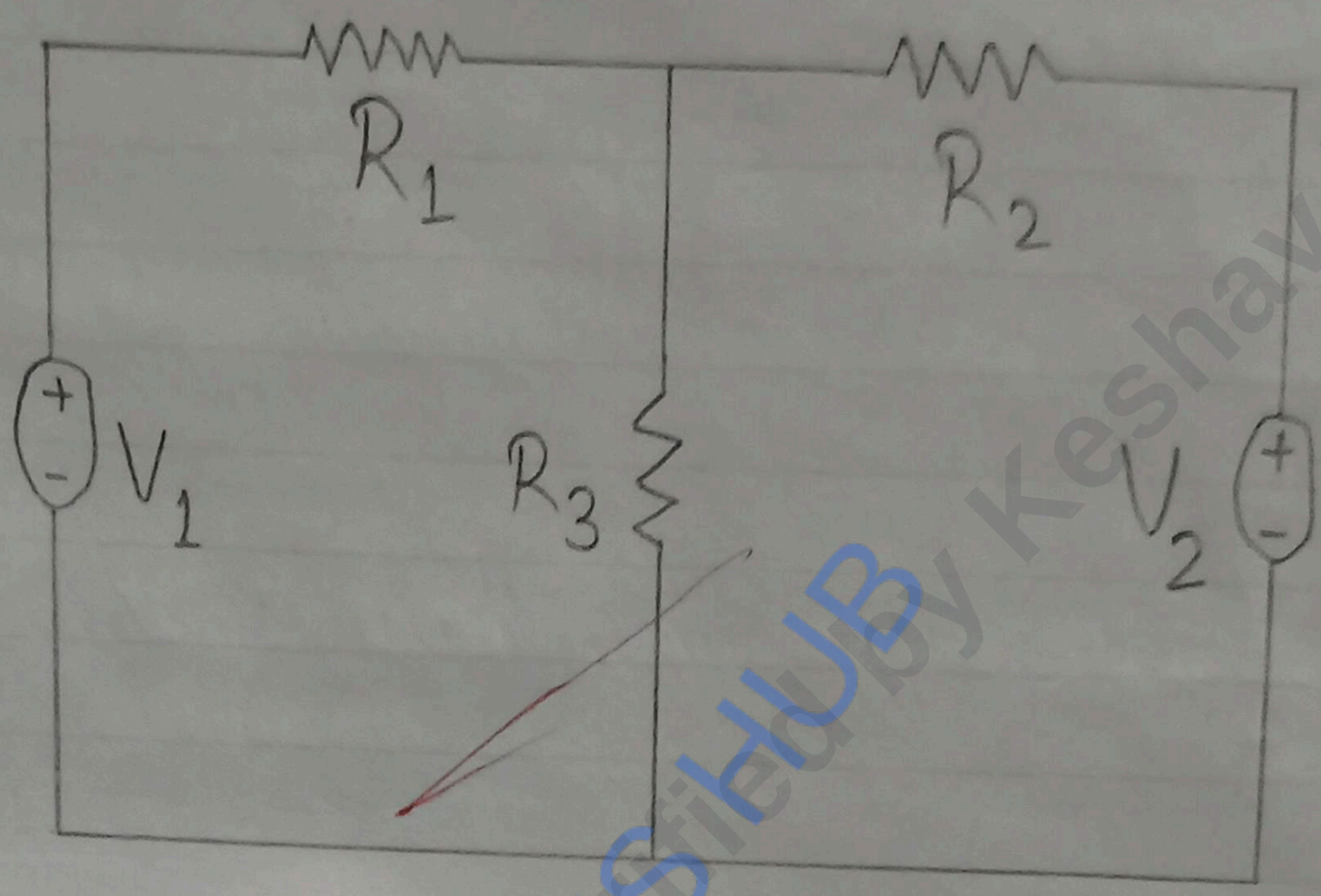
Name of the Experiment	Page No.	Date of Experiment	Date of Submission	Remarks
Verification of superposition	2-4	08/10/26	10/10/26	

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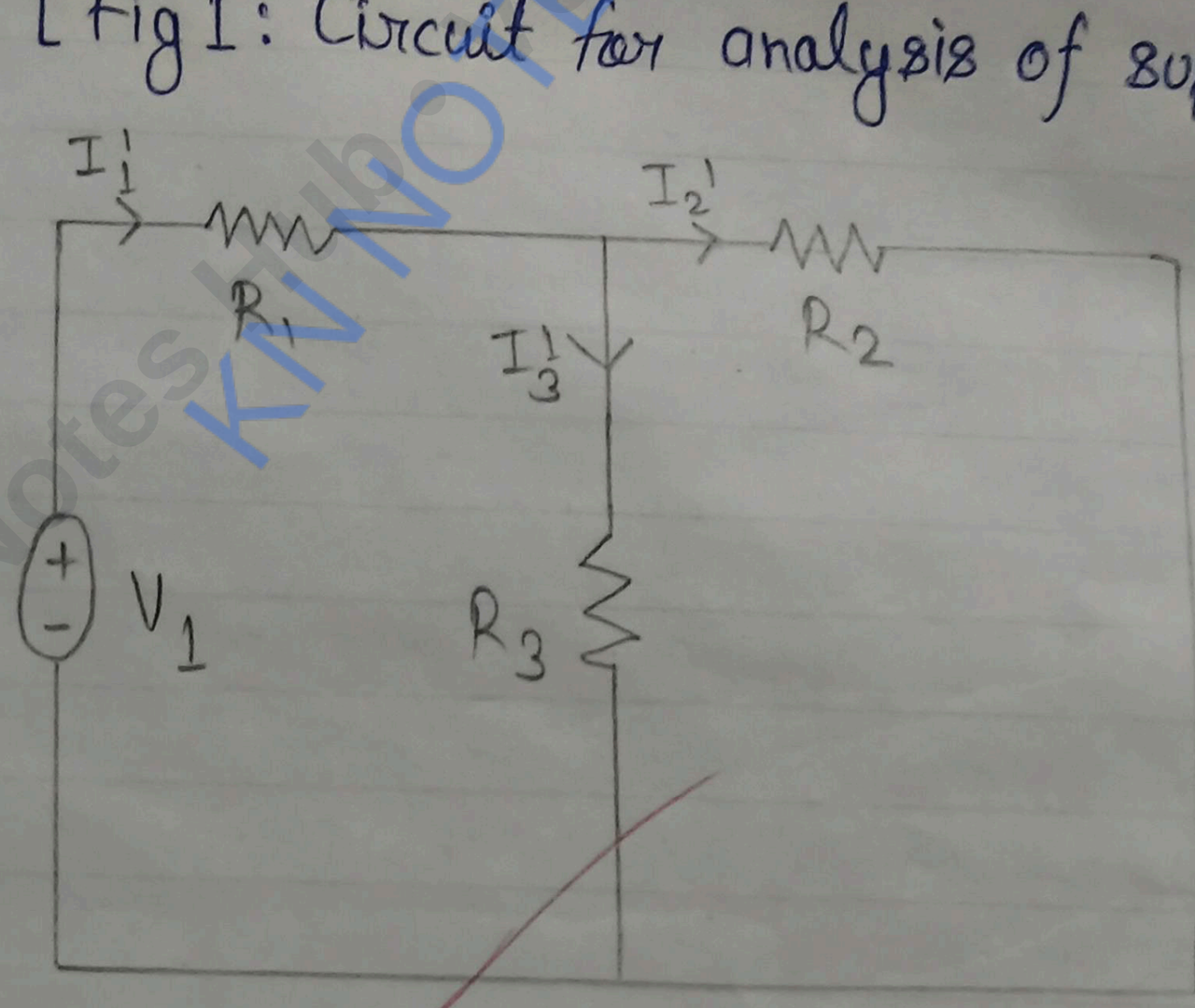
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Circuit Diagram :→



[fig 1: Circuit for analysis of superposition th~



[fig 2: Circuit with only V_2 short circuit]

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Experiment → 1

* Objective → To verify superposition theorem.

→ Apparatus :→ Virtual Lab.

→ Theory →

If a number of voltage or current source are simulataneously in a linear network the resultant current in any branch is the algebric sum of the currents that would be produced in it, when each source acts alone replacing all other independent sources by their internal resistances.

In given figure I apply superposition theorem, let us first take the sources V_1 alone at first replacing V_2 by short circuit as shown in figure 2. Here;

$$I_1' = \frac{V_1}{\frac{R_2 \times R_3}{R_2 + R_3} + R_1}$$

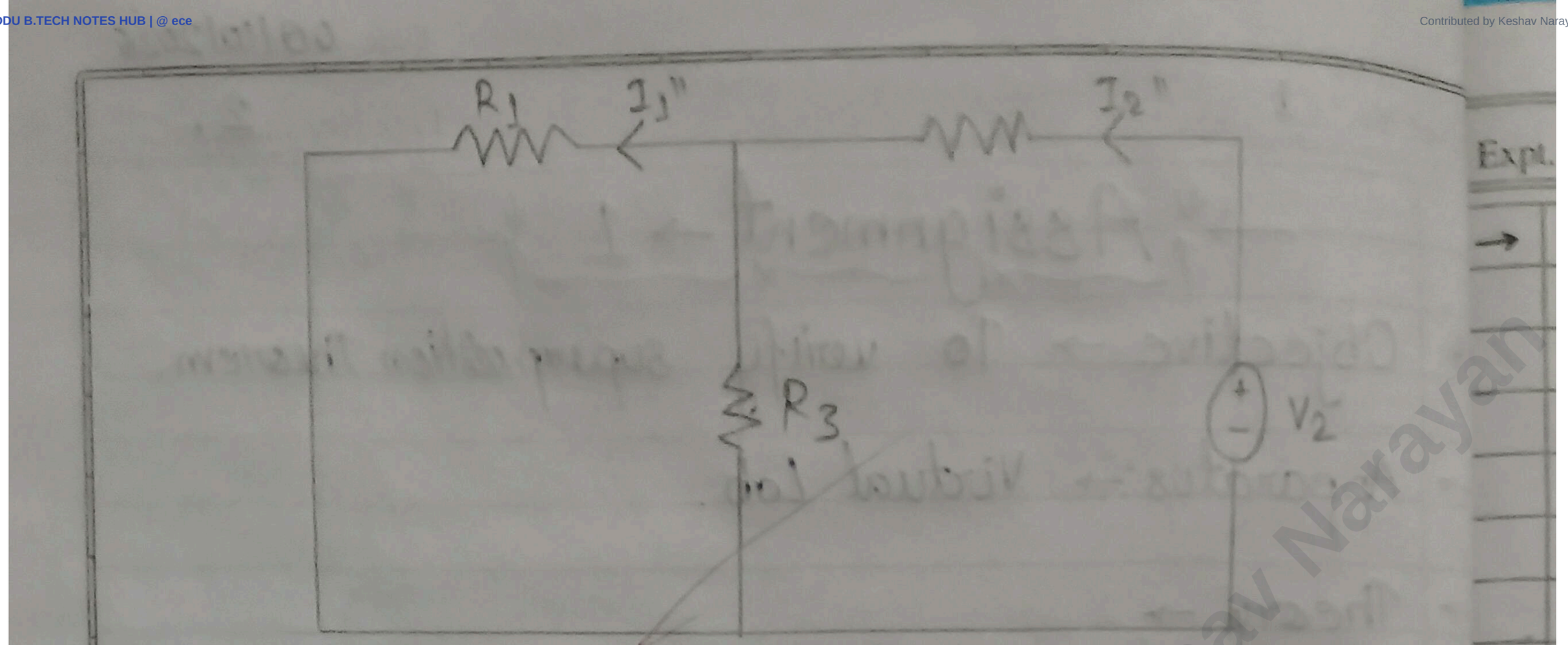
$$I_2' = I_1' \times \frac{R_3}{R_2 + R_3}$$

$$I_3' = I_1' - I_2'$$

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[fig 3: Cuircuit with only V1 short cuircut]

Calculation :→

Case 1 →

for I_1,

$I_1 = I_1' - I_1''$
$= 0.56770 - 0.17740$

$[I_1 = 0.39032 \text{ Amp}]$

for I_2,

$I_2 = I_2' - I_2''$
$= 0.35484 - 0.24839$
$I_2 = 0.10645$ Amp

Since the actual dir^n is opposite to the reference dir^n,

$[I_2 = -0.10645 \text{ Amp}]$

for I_3,

$I_3 = I_3' + I_3''$
$= 0.21290 + 0.70968$
$[I_3 = 0.28387 \text{ Amp}]$

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Next, removing V_1 by short circuit, let the circuit be energized by V_2 only as shown in figure 3.

Then,

$$I_2'' = \frac{V_2}{\frac{R_1 \times R_3}{R_1 + R_3} + R_2}$$

$$I_1'' = I_2'' \times \frac{R_3}{R_1 + R_3}$$

$$I_3'' = I_2'' - I_1''$$

As per superposition theorem;

$$I_3 = I_3' + I_3''$$

$$I_2 = I_2 - I_2''$$

$$I_1 = I_1' - I_1''$$

* Observation Table →

no.	In presence of both V_1 and V_2			presence of V_1 only			presence of V_2		
	Branch current I_1 (Amp)	I_2 (Amp)	I_3 (Amp)	I_1 (Amp)	I_2 (Amp)		I_1 (amp)	I_2 (amp)	(I_3)
t	0.39032	-0.10645	0.28387	0.5677	0.35484	0.2129	-0.1774	0.24839	0.70968

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Case 2 →

for I_1',

$$I_1' = \frac{V_1}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}$$

$$I_1' = \frac{220}{200 + \frac{300 \cdot 500}{300 + 500}} \Rightarrow \frac{220}{200 + 187.5}$$

$$\boxed{I_1' = 0.56774\ A}$$

for I_2',

$$I_2' = I_1' \frac{R_3}{R_2 + R_3}$$

$$= 0.56774 \times \frac{500}{300 + 500}$$

$$\boxed{I_2' = -0.35484\ A}$$

for I_3',

$$I_3' = I_1' - I_2'$$

$$= 0.56774 - 0.3548$$

$$\boxed{I_3' = 0.21290\ A}$$

Case 3 :→

for I_1'',

$$I_1'' = I_2'' \frac{R_3}{R_1 + R_3}$$

$$= 0.24839 \times \frac{500}{200 + 500}$$

$$\boxed{I_1'' = -0.17742\ A}$$

for I_2'',

$$I_2'' = \frac{V_2}{R_2 + \frac{R_1 R_3}{R_1 + R_3}}$$

$$= \frac{110}{300 + \frac{200 \times 500}{200 + 500}} = \frac{110}{300 + 142}$$

$$\boxed{I_2'' = 0.24839\ A}$$

for I_3'',

$$I_3'' = I_2'' - I_1''$$

$$= 0.24839 - 0.17742$$

$$\boxed{I_3'' = 0.07097\ A}$$

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* Result →

The superposition theorem is successfully verified because the currents obtained when both sources act together are approximately equal to the algebric sum of the currents produced by each source acting separately.

Case ① I_1 = Case ② I_1' + Case ③ I_1''

$0.3903 = 0.3903$

Case ① I_2 = Case ② I_2' + Case ③ I_2''

$-0.10645 = -0.10645$

Case ① I_3 = Case ② I_3' + Case ③ I_3''

$0.28387 = 0.28387$

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